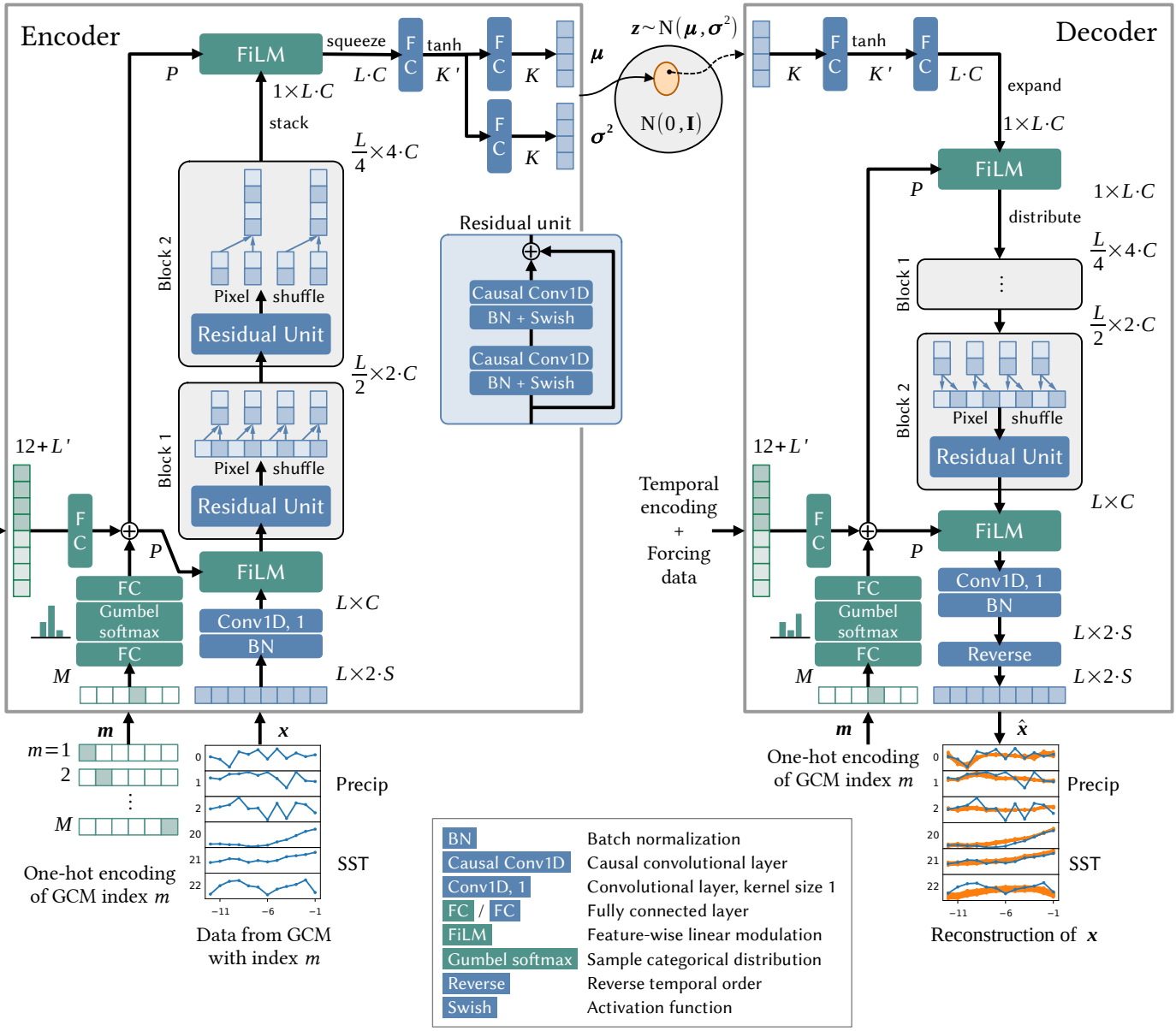
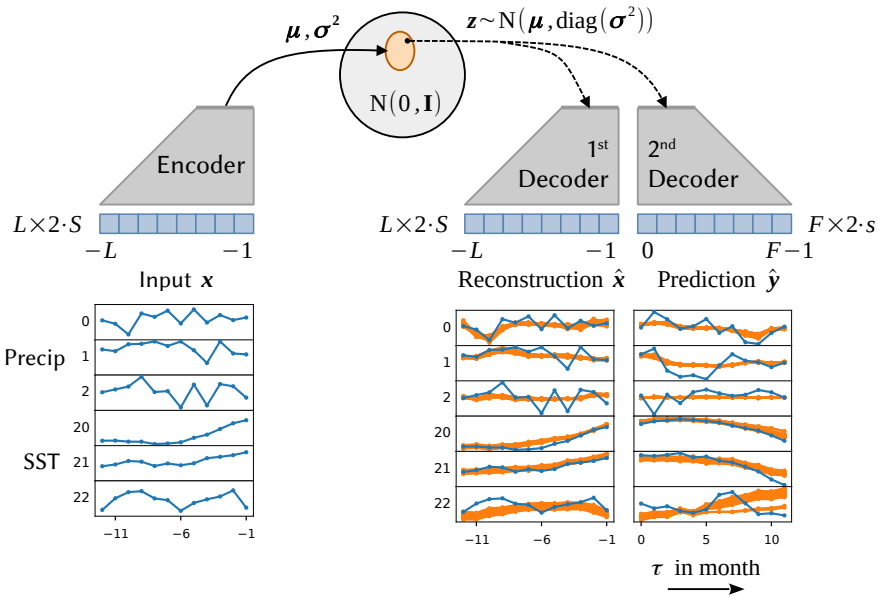

Encoder
Decoder
FiLM
squeeze
tanh
FC
stack
expand
distribute
Pixel shuffle
Residual Unit
Residual unit
Causal Conv1D
BN + Swish
Block 1
Block 2
Temporal encoding + Forcing data
Gumbel softmax
Conv1D, 1
BN
Reverse
$z \sim N(\mu, \sigma^2)$
$N(0, \mathbf{I})$
μ
σ^2
$12+L'$
$1\times L\cdot C$
$\frac{L}{4}\times 4\cdot C$
$\frac{L}{2}\times 2\cdot C$
$L\times C$
$L\times 2\cdot S$
One-hot encoding of GCM index m
Data from GCM with index m
Reconstruction of x
Precip
SST
BN Batch normalization
Causal Conv1D Causal convolutional layer
Conv1D, 1 Convolutional layer, kernel size 1
FC / FC Fully connected layer
FiLM Feature-wise linear modulation
Gumbel softmax Sample categorical distribution
Reverse Reverse temporal order
Swish Activation function


$\boldsymbol{\mu}, \boldsymbol{\sigma}^2$
$\boldsymbol{z} \sim \mathrm{N}(\boldsymbol{\mu}, \mathrm{diag}(\boldsymbol{\sigma}^2))$
$\mathrm{N}(0, \mathbf{I})$
Encoder
1st Decoder
2nd Decoder
$L \times 2 \cdot S$
$-L$
-1
Input $\boldsymbol{x}$
$L \times 2 \cdot S$
$-L$
-1
Reconstruction $\hat{\boldsymbol{x}}$
0
$F-1$
Prediction $\hat{\boldsymbol{y}}$
$F \times 2 \cdot s$
Precip
SST
0
1
2
20
21
22
−11
−6
−1
0
5
10
τ in month